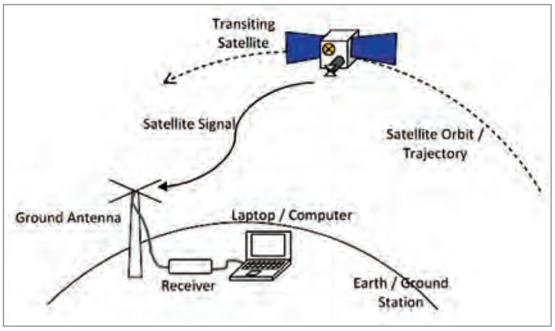

Fig. 3: Basic system requirements

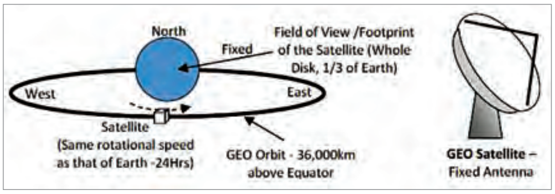

Fig. 4: Geosynchronous satellite orbit and antenna position

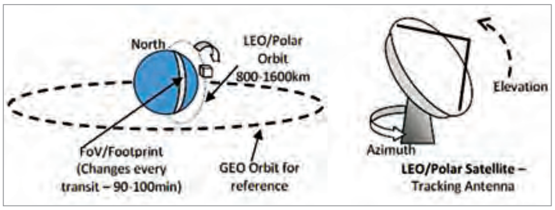

Fig. 5: Polar orbit and tracking antenna

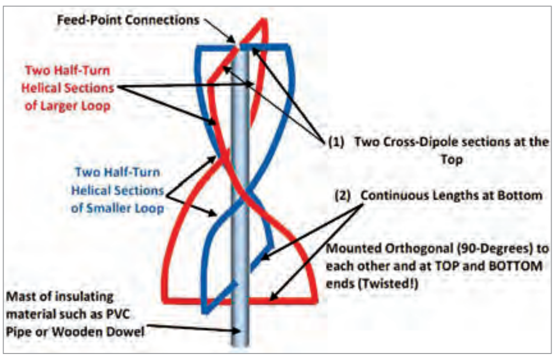

Fig. 6: The quadrifilar helical (QFH) antenna

137MHz band. All image data from the infrared and near-visible range imagers on-board these satellites is transmitted along with the necessary telemetry and sync signals in the same audio signal, and this is termed as Analogue Transmission. We shall receive the weather images from these satellites for this project.

A satellite from Russia, named Meteor-M-2, in the polar orbit transmits weather images at even higher resolution in the same 137MHz band. But it uses digital QPSK modulation, so requires a special demodulator to reconstruct the images from the digital data transmission.

Practical system build-up

Antenna. As stated earlier, the satellites we are interested to receive are the NOAA-15, NOAA-18, and the NOAA-19 that transmit the images on 137MHz VHF band with a frequency modulated (FM) audio information using the APT protocol. Details about this protocol are given in later part of the article.

NOAA-15 transmits at 137.6200MHz, NOAA-18 at 137.9125MHz, and NOAA-19 at 137.100MHz with a carrier deviation of ±17kHz. Thus, a receiver of at least 40kHz bandwidth is needed. Also, this transmission uses the right-hand-circular-polarisation (RHCP) at the satellite antenna. Eventually, you shall theoretically require a circularly polarised antenna at the receiving end too. Alternately, if you use a linearly polarised antenna, only half the power is received because only one component, horizontal or vertical, is sensed by the linear antenna.

The QFH antenna. Quadrifilar Helix Antenna, or simply QFH antenna, is the right choice for circularly polarised signal. But

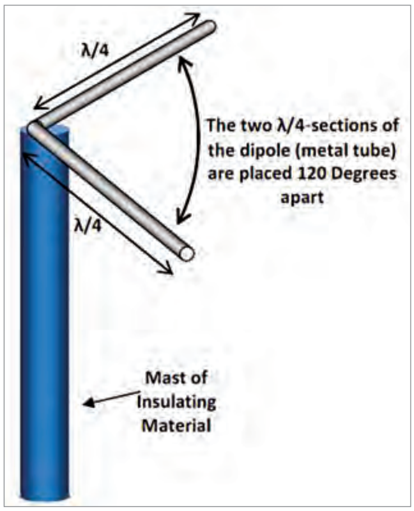

Fig. 8: The Vee-dipole antenna

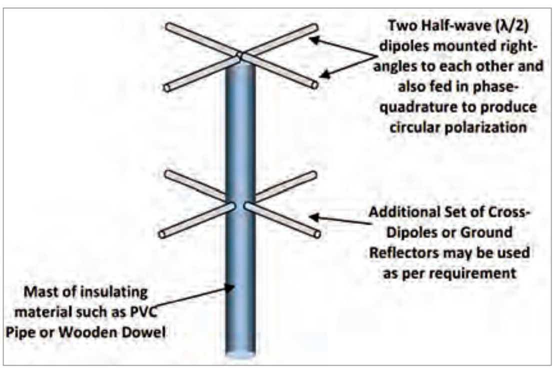

Fig. 7: The turnstile antenna

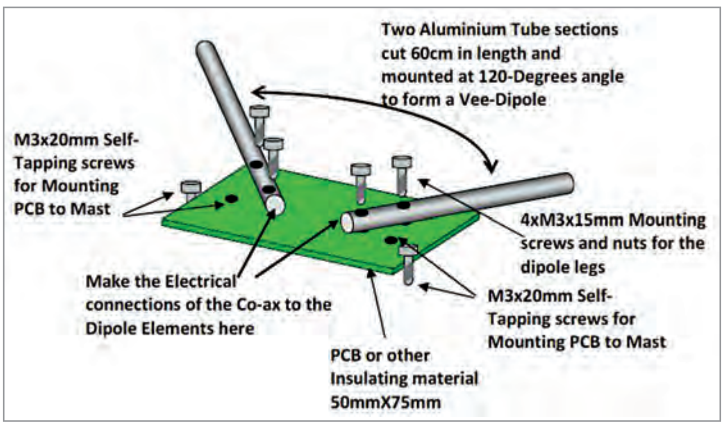

Fig. 9: The mounting and mechanical assembly details of the Vee-dipole